

02477 – Bayesian Machine Learning: Lecture 3

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Outline

- 1 Recap: Probabilistic machine learning
- 2 Recap: Multivariate Gaussian distributions
- 3 Recap: Linear regression and supervised learning
- 4 Bayesian linear regression
- 5 The posterior predictive distribution
- 6 Dealing with hyperparameters

Recap: Probabilistic machine learning

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

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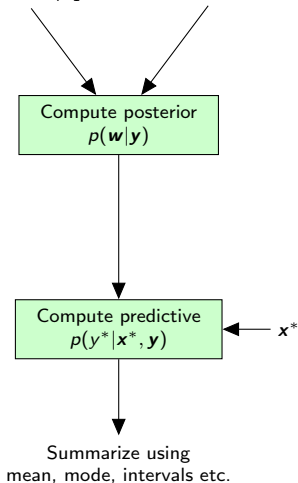
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- Key take-away:** To reason about y^* *given* y , we need to *average the likelihood* for y^* wrt. to the *posterior distribution* $p(\theta|y)$.

Bayesian inference for supervised learning

$$\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N \quad p(\mathbf{w}, \mathbf{y}) = p(\mathbf{y}|\mathbf{w})p(\mathbf{w})$$



- Same principles for linear regression, logistic regression, neural networks etc. etc.

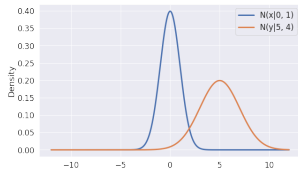
Recap: Multivariate Gaussian distributions

Univariate normal distribution

- The *normal distribution* (also known as the Gaussian)

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Two parameters: $\mu = \mathbb{E}[x]$ and $\sigma^2 = \mathbb{V}[x]$
- Widely due to Central limit theorems, maximum entropy principle, relation to least squares minimization, nice mathematical properties



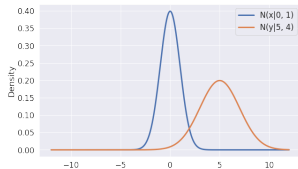
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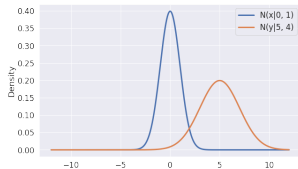
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- Let $x \sim \mathcal{N}(m_x, v_x)$ and $y \sim \mathcal{N}(m_y, v_y)$ for $x \perp y$, then

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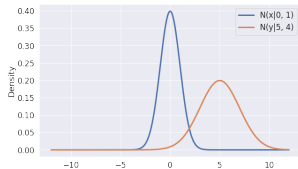
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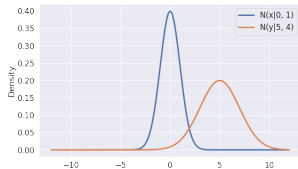
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- Functional form: The *logarithm of a Gaussian density* is a *second order polynomial*

$$\ln \mathcal{N}(x|\mu, \sigma^2) = -\frac{1}{2\sigma^2}x^2 + \frac{\mu}{\sigma^2}x + K$$

The functional form of a Gaussian distribution I

- Recall we discussed the *functional form* of a Beta distribution

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- Let's derive the functional form of the Gaussian

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The functional form of a Gaussian distribution II: example

- Take-away: the *the functional form* of a univariate Gaussian is

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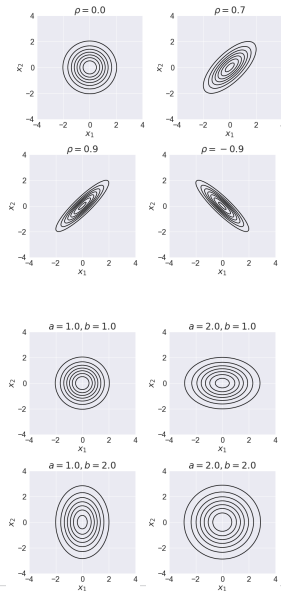
- Therefore, we conclude $p(x) = \mathcal{N}(x|4, 2)$.

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■ Two parameters: $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$ and $\boldsymbol{\Sigma} = \text{cov}[\mathbf{x}]$



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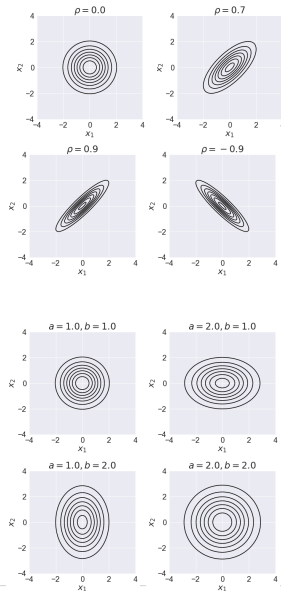
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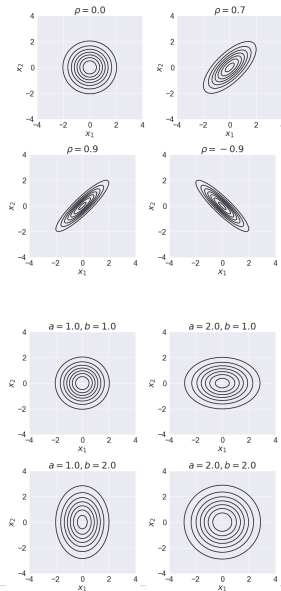
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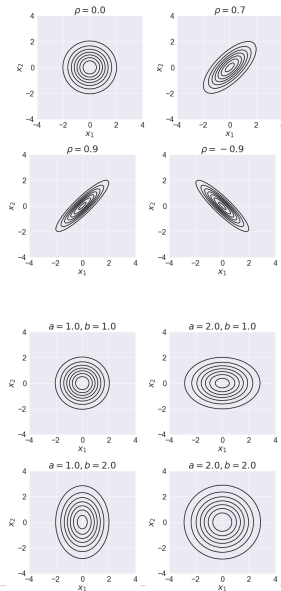
■ Two parameters: $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$ and $\boldsymbol{\Sigma} = \text{cov}[\mathbf{x}]$

■ Covariance matrix for $D = 2$

$$\boldsymbol{\Sigma} = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$$

■ Correlation coefficient $\rho = \frac{c}{\sqrt{ab}}$

■ Let $\mathbf{x} \sim \mathcal{N}(\mathbf{m}_x, \mathbf{V}_x)$ and $\mathbf{y} \sim \mathcal{N}(\mathbf{m}_y, \mathbf{V}_y)$ for $\mathbf{x} \perp \mathbf{y}$, then



The multivariate normal distribution

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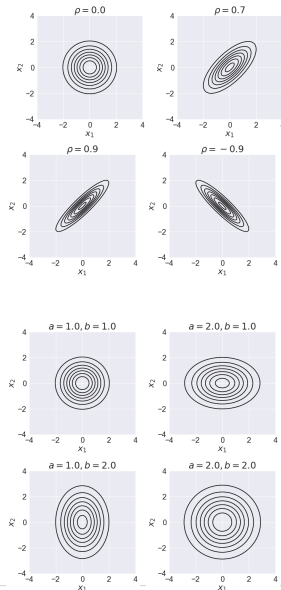
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$$\mathbf{a} + \mathbf{B}\mathbf{x} \sim \mathcal{N}(\mathbf{a} + \mathbf{B}\mathbf{m}_x, \mathbf{B}\mathbf{V}_x\mathbf{B}^T)$$



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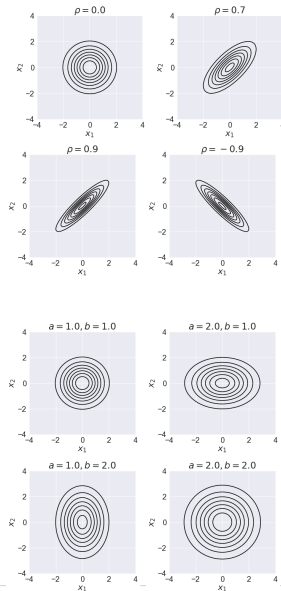
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$$\mathbf{x} + \mathbf{y} \sim \mathcal{N}(\mathbf{m}_x + \mathbf{m}_y, \mathbf{V}_x + \mathbf{V}_y)$$



The functional form of multivariate Gaussians

- Consider now the log density, focussing only on terms dependent on $\mathbf{x}$.

$$\ln p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \ln \left[(2\pi)^{-\frac{D}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right) \right]$$

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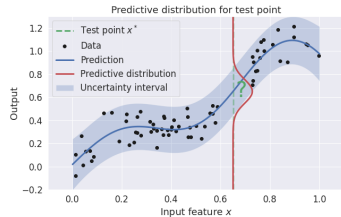
Key take-aways

- Every time we encounter a distribution with a *quadratic log density*, it must be a Gaussian distribution (if $\boldsymbol{\Sigma}$ is a valid covariance matrix)
- We can *match coefficients* of first and second order term to *determine mean and covariance*

Recap: Linear regression and supervised learning

Supervised learning: linear regression

- Dataset $\mathcal{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\}$
 - Input features: $\mathbf{x}_i \in \mathbb{R}^D$
 - Targets: $y_i \in \mathbb{R}$



Supervised learning: linear regression

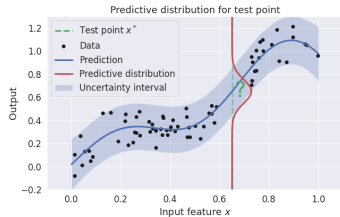
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■ Input features: $\mathbf{x}_i \in \mathbb{R}^D$

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■ Additive noise models

$$y_i = f(\mathbf{x}_i | \mathbf{w}) + \epsilon_i$$



Supervised learning: linear regression

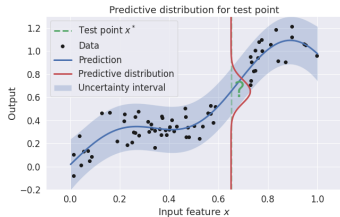
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- Linear models are *linear wrt. parameters*, not data!

$$f(\mathbf{x} | \mathbf{w}) = w_0 + w_1 x_1 + \dots + w_M x_M = \mathbf{w}^T \mathbf{x}$$



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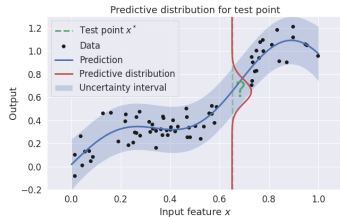
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$$f(\mathbf{x} | \mathbf{w}) = w_0 + w_1 x_1 + \dots + w_M x_M = \mathbf{w}^T \mathbf{x}$$

- *Non-linear* feature extractors $\phi(\cdot)$ (basis functions)

$$f(\mathbf{x} | \mathbf{w}) = \sum_{j=0}^M w_j \phi_j(\mathbf{x}) = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x})$$



Linear regression: the probabilistic model

- The *predictive distribution* of y^* given $\mathbf{x}^*$ and the data $\mathcal{D}$ is our goal

$$p(y^*|\mathcal{D}, \mathbf{x}^*)$$

- Model for the "signal"

$$f(\mathbf{x}_i|\mathbf{w}) = \mathbf{w}^T \phi(\mathbf{x}_i)$$

- The gaussian noise ϵ_i is assumed to be *independent and identically distributed (i.i.d)*

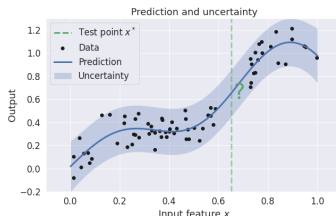
$$y_i = f(\mathbf{x}_i|\mathbf{w}) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- The *likelihood* for the i 'th data point

$$p(y_i|\mathbf{x}_i, \mathbf{w}, \sigma^2) = \mathcal{N}(y_i|\mathbf{w}^T \phi(\mathbf{x}_i), \sigma^2)$$

- Using the maximum likelihood solution as a *plug-in* estimator

$$p(y^*|\mathcal{D}, \mathbf{x}^*) = \mathcal{N}(y_i|\hat{\mathbf{w}}_{\text{MLE}}^T \phi(\mathbf{x}_*), \sigma^2)$$



Estimating the parameters using maximum likelihood

- Given a dataset $\mathcal{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\}$, the *likelihood* for dataset is

$$p(\mathbf{y}|\mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(y_n | f(\mathbf{x}_n | \mathbf{w}), \sigma^2)$$

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- Taking the logarithm and using $f(\mathbf{x}_n, \mathbf{w}) = \mathbf{w}^T \phi(\mathbf{x}_n)$

$$\ln p(\mathbf{y}|\mathbf{w}, \sigma^2) = \sum_{n=1}^N \ln \mathcal{N}(y_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

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- Maximum likelihood estimator $\hat{\mathbf{w}}_{\text{MLE}}$ is equivalent to minimizing sum-of-squares error

$$\hat{\mathbf{w}}_{\text{MLE}} = \left(\Phi^T \Phi \right)^{-1} \Phi^T \mathbf{y} \quad (\text{Normal equations})$$

Estimating the parameters using maximum likelihood

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$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{\mathbf{w}}_{\text{MLE}}^T \phi(\mathbf{x}_n))^2$$

Example: Polynomial regression using maximum likelihood I

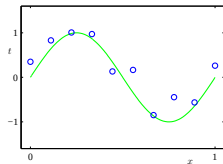
Example from Bishop

■ Polynomial basis functions

$$f(x|\mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \mathbf{w}^T \phi(x)$$

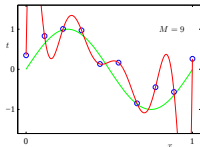
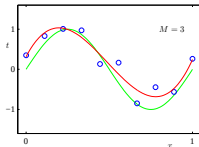
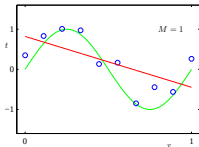
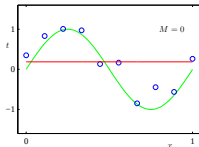
■ Feature transformations

$$\phi(x) = [1 \quad x \quad x^2 \quad \dots \quad x^M]^T$$



■ M controls the *model complexity*: Underfitting vs overfitting

■ *Model selection*: How to choose M ?



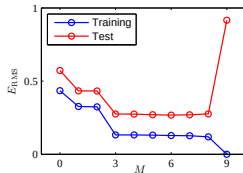
Example: Polynomial regression using maximum likelihood II

■ Cross-validation

Split data into training and test sets

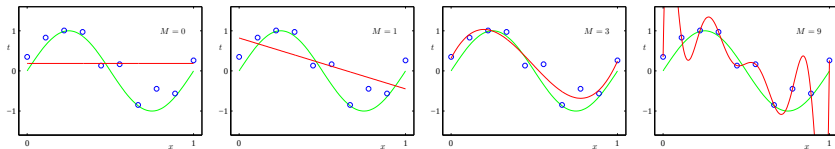
■ Overfitting (low training error, high test error)

As the function become more flexible we start to fit the noise in the data



■ "Underfitting" (high training error, high test error)

When the function is not sufficiently flexible to fit the data



Example: Polynomial regression using maximum likelihood III

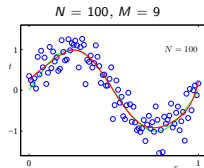
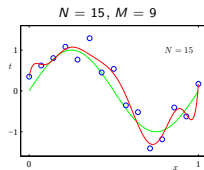
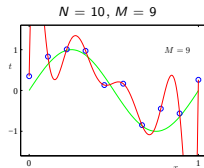
- *The optimal model complexity depends on the amount of data*

The more data the more flexible model we can "afford" to fit

- *Regularization: Controlling the effective model complexity*

We can use regularization to control the effect model complexity when we have limited data

	$M = 0$	$M = 1$	$M = 6$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43



Regularized least squares

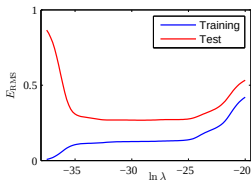
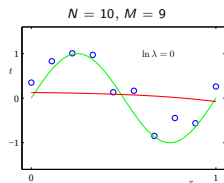
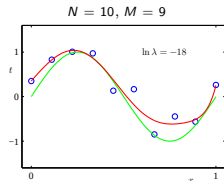
- Recall: *Maximum likelihood* is equivalent *minimizing sum-of-squares error*

$$E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y_n - f(x_n|\mathbf{w}))^2$$

- Adding *penalty term* to prevent weights from becoming too large

$$\tilde{E}_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y_n - f(x_n|\mathbf{w}))^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

- What happens when $\lambda = 0$? $\lambda \rightarrow \infty$?
- Many names: Ridge regression, shrinkage, weight decay
- Regularization parameter λ controls the *effective complexity*



Bayesian linear regression

Bayesian Linear regression: motivation

■ *Overfitting*

Maximum likelihood can be problematic for flexible models

■ *Controlling model complexity*

Limiting number of basis functions and/or regularization?

■ *Model selection*

How to choose the optimal value of λ ?

■ *Cross-validation*

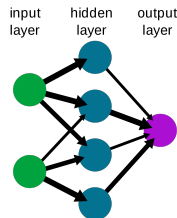
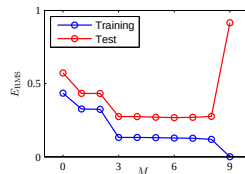
Training + validation/development + test

■ *Bayesian methods*

- less prone to overfitting
- can (often) adapt model complexity automatically

■ *Applications in modern machine learning*

1. Small datasets
2. Transfer learning
3. Component in more complex models
4. Simple uncertainty quantification for neural networks



Bayesian Linear regression: prior and likelihood

- Simplified set-up: assuming σ^2 is fixed and known, then Bayes' rule states

$$p(\mathbf{w}|\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{w})p(\mathbf{w})}{p(\mathbf{y})}$$

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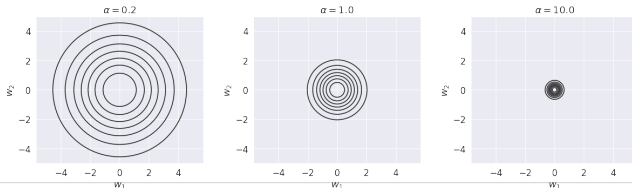
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- The *posterior distribution* of the weights $\mathbf{w}$ given the data $\mathbf{y}$ is given by

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- The *mode of the posterior (MAP)* is equivalent to ridge regression with $\lambda = \frac{\alpha}{\beta}$ and to maximum likelihood when $\alpha \rightarrow 0$

Deriving the posterior distribution of the weights

Recall the functional form of a generic multivariate Gaussian $\mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S})$

$$\ln \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}) = -\frac{1}{2} \mathbf{w}^T \mathbf{S}^{-1} \mathbf{w} + \mathbf{m}^T \mathbf{S}^{-1} \mathbf{w} + \text{constant}$$

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$$\mathbf{S}^{-1} = \beta \Phi^T \Phi + \alpha \mathbf{I} \quad \Longleftrightarrow \quad \mathbf{S} = (\beta \Phi^T \Phi + \alpha \mathbf{I})^{-1}$$

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- Equating coefficients for the first order term

$$\mathbf{m}^T \mathbf{S}^{-1} = \beta \mathbf{y}^T \Phi \quad \Longleftrightarrow \quad \mathbf{m} = \beta \mathbf{S} \Phi^T \mathbf{y}$$

Bayesian linear regression model: the key equations

- Given design matrix $\Phi \in \mathbb{R}^{N \times D}$ and observations $\mathbf{y} \in \mathbb{R}^N$:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}) \quad (\text{prior})$$

$$p(\mathbf{y} | \mathbf{w}) = \mathcal{N}(\mathbf{y} | \Phi \mathbf{w}, \sigma^2 \mathbf{I}) \quad (\text{likelihood})$$

$$p(\mathbf{w} | \mathbf{y}) = \mathcal{N}(\mathbf{w} | \mathbf{m}, \mathbf{S}) \quad (\text{posterior})$$

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y} | \mathbf{0}, \sigma^2 \mathbf{I} + \alpha^{-1} \Phi \Phi^T) \quad (\text{marginal likelihood})$$

- The *posterior parameters* are given by (using $\beta \equiv \frac{1}{\sigma^2}$)

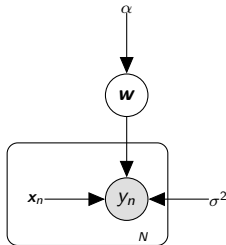
$$\mathbf{m} = \beta \mathbf{S} \Phi^T \mathbf{y}$$

$$\mathbf{S} = \left(\alpha \mathbf{I} + \beta \Phi^T \Phi \right)^{-1}$$

- Two hyperparameters*

α : prior precision of the regression weights

β : precision of the measurements



Linear Gaussian-systems in general (see Section 3.3 in Murphy1)

- For *linear* systems: the Gaussian distribution is *conjugate* to itself
- The *posterior* for a *linear* Gaussian model with Gaussian prior is also *Gaussian*

$$p(\mathbf{y}|\mathbf{z}) = \mathcal{N}(\mathbf{y}|\mathbf{W}\mathbf{z} + \mathbf{b}, \Sigma_y) \qquad p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_z, \Sigma_z)$$

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$$p(\mathbf{y}|\mathbf{z}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_{z|y}, \Sigma_{z|y})$$

$$\Sigma_{z|y}^{-1} = \Sigma_z^{-1} + \mathbf{W}^T \Sigma_y^{-1} \mathbf{W}$$

$$\boldsymbol{\mu}_{z|y} = \Sigma_{z|y} \left[\mathbf{W}^T \Sigma_y^{-1} (\mathbf{y} - \mathbf{b}) + \Sigma_z^{-1} \boldsymbol{\mu}_z \right]$$

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$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_z \\ \mathbf{W}\boldsymbol{\mu}_z + \mathbf{b} \end{bmatrix} \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_z & \Sigma_z \mathbf{W}^T \\ \mathbf{W}\Sigma_z & \Sigma_y + \mathbf{W}\Sigma_z \mathbf{W}^T \end{bmatrix}$$

- The *posterior* distribution of $\mathbf{z}$ given $\mathbf{y}$

$$\begin{aligned} p(\mathbf{y}|\mathbf{z}) &= \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_{z|y}, \Sigma_{z|y}) \\ \Sigma_{z|y}^{-1} &= \Sigma_z^{-1} + \mathbf{W}^T \Sigma_y^{-1} \mathbf{W} \\ \boldsymbol{\mu}_{z|y} &= \Sigma_{z|y} \left[\mathbf{W}^T \Sigma_y^{-1} (\mathbf{y} - \mathbf{b}) + \Sigma_z^{-1} \boldsymbol{\mu}_z \right] \end{aligned}$$

- The *marginal* distribution $\mathbf{y}$

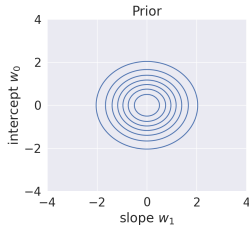
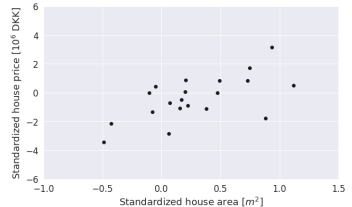
$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y}|\mathbf{W}\boldsymbol{\mu}_z + \mathbf{b}, \Sigma_y + \mathbf{W}\Sigma_z \mathbf{W}^T)$$

Example

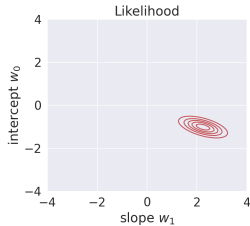
- Simple linear model for fictive house prices

$$f(x|\mathbf{w}) = w_0 + w_1 x$$

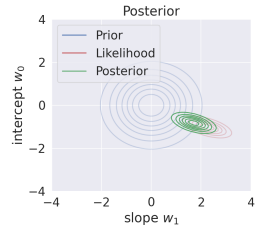
- The posterior summarizes our beliefs about the parameters after seeing the data



$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} | 0, \alpha^{-1} \mathbf{I})$$



$$p(\mathbf{y} | \mathbf{w}) = \mathcal{N}(\mathbf{y} | \Phi \mathbf{w}, \sigma^2 \mathbf{I})$$



$$p(\mathbf{w} | \mathbf{y}) = \mathcal{N}(\mathbf{w} | \mathbf{m}, \mathbf{S})$$

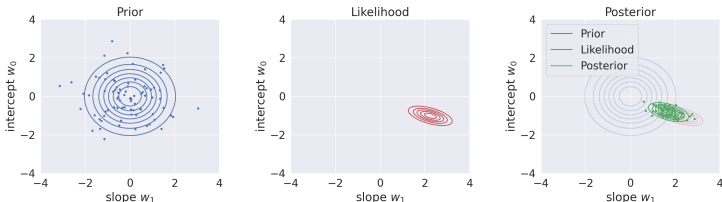
Posterior inference

$$\text{house price} = w_0 + w_1 \cdot \text{area}$$

- Question: Are larger areas associated with bigger house prices?
- We can calculate various probabilities of interest directly from the posterior (analytically or via sampling), e.g.

$$p(w_1 > 0 | y) \approx 0.99$$

- No need to remember whether to should use t-tests, F-tests, χ^2 -tests etc



The posterior predictive distribution

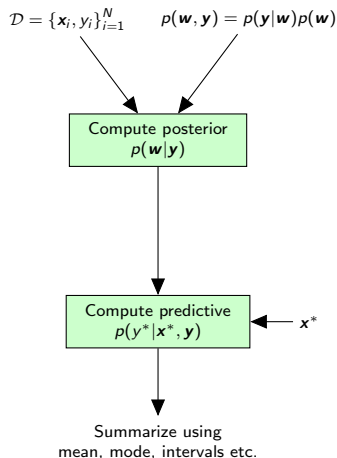
But how about making predictions?

- We defined a *linear* regression model

$$y_i = \mathbf{w}^T \phi(\mathbf{x}_i) + \epsilon_i$$

- We derived the *posterior distribution*

$$p(\mathbf{w}|\mathbf{y}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S})$$



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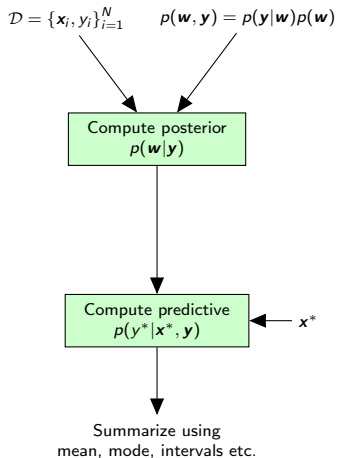
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$$p(y^*|\mathbf{x}^*, \mathbf{w}) = \mathcal{N}(y^*|\mathbf{w}^T \underbrace{\phi(\mathbf{x}^*)}_{\phi_*}, \sigma^2)$$



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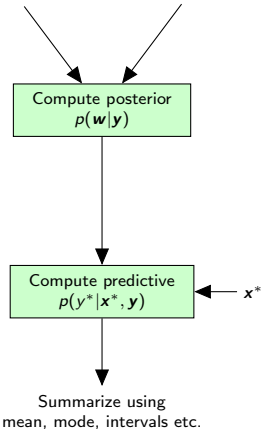
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- The *posterior predictive distribution* is given by

$$p(y^*|\mathbf{y}, \mathbf{x}^*) = \int \underbrace{p(y^*|\mathbf{x}^*, \mathbf{w})}_{\text{Pred. likelihood}} \underbrace{p(\mathbf{w}|\mathbf{y})}_{\text{Posterior}} d\mathbf{w}$$

$$\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N \quad p(\mathbf{w}, \mathbf{y}) = p(\mathbf{y}|\mathbf{w})p(\mathbf{w})$$



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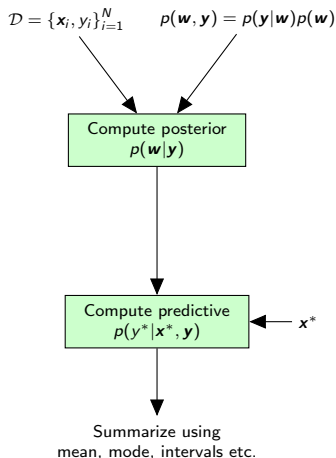
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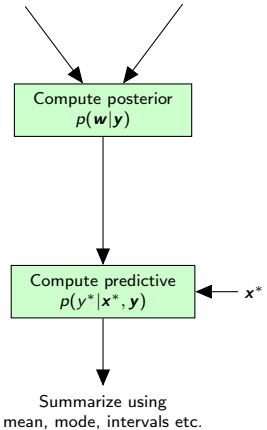
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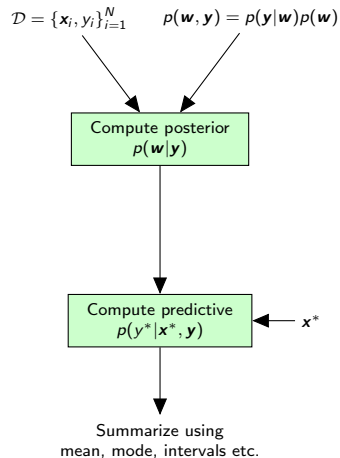
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- Can be derived using eq. (3.38) in Murphy1



Quiz time!

$$p(\mathbf{w}|\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{w})p(\mathbf{w})}{p(\mathbf{y})} \quad (\textit{Bayes' rule})$$

$$p(\mathbf{y}) = \int p(\mathbf{y}|\mathbf{w})p(\mathbf{w})d\mathbf{w} \quad (\textit{marginal likelihood})$$

$$p(y^*|\mathbf{y}, \mathbf{x}^*) = \int p(y^*|\mathbf{x}^*, \mathbf{w})p(\mathbf{w}|\mathbf{y})d\mathbf{w} \quad (\textit{Posterior predictive dist.})$$

- Spend 5 minutes DTU Learn quiz: "Lecture 3: Bayesian inference"

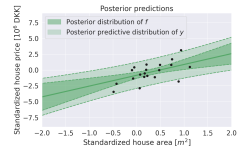
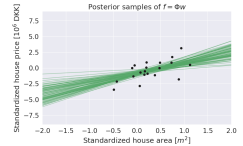
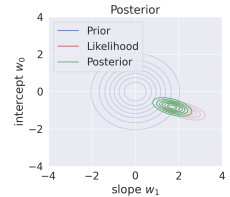
Posterior Predictive distributions

Building intuition

- The *posterior distribution* is $p(\mathbf{w}|\mathbf{y}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S})$
- Two *model quantities* of interest

$$f^* = f(\mathbf{x}_*) = \mathbf{w}^T \boldsymbol{\phi}_*$$

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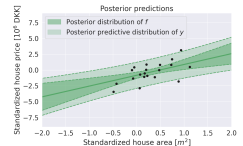
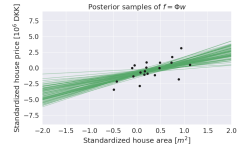
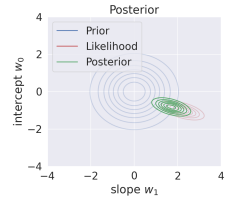
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■ Computing the *posterior* mean and variance

$$\mathbb{E}[f^*] =$$

$$\mathbb{V}[f^*] =$$



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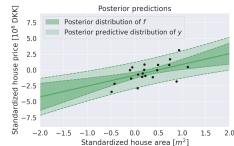
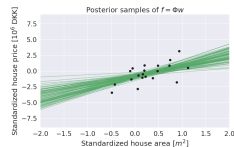
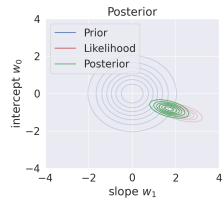
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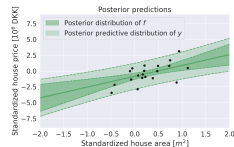
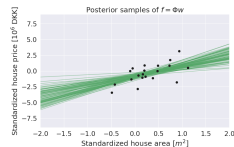
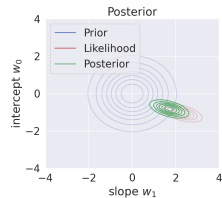
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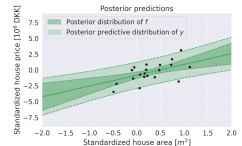
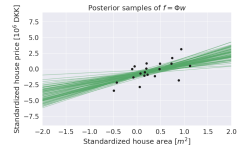
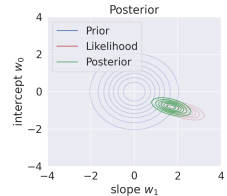
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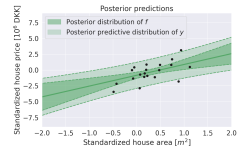
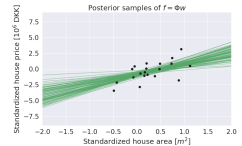
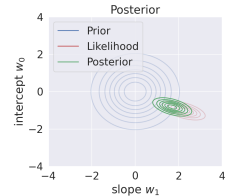
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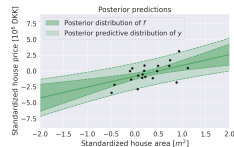
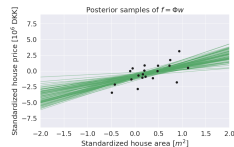
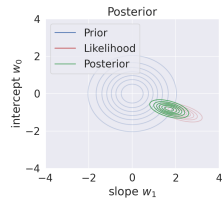
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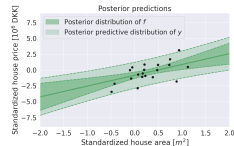
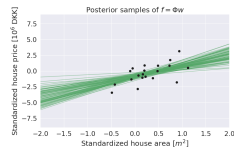
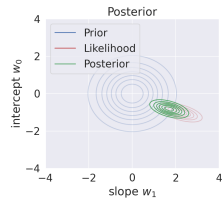
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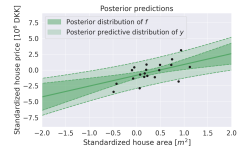
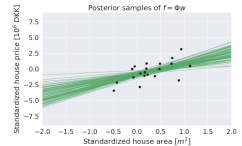
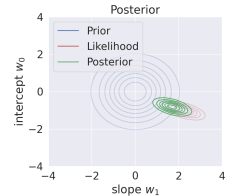
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- Hence, the *posterior distribution* of f^* given the data $\mathbf{y}$ is

$$p(f^*|\mathbf{y}, \mathbf{x}_*) = \mathcal{N}(f^*|\mathbf{m}^T \boldsymbol{\phi}_*, \boldsymbol{\phi}_*^T \mathbf{S} \boldsymbol{\phi}_*)$$



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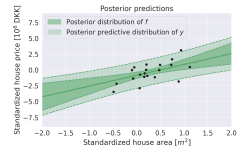
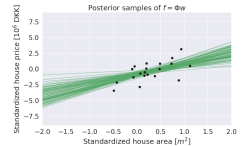
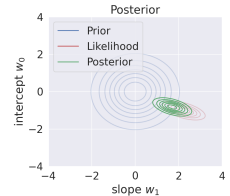
- Hence, the *posterior distribution* of f^* given the data $\mathbf{y}$ is

$$p(f^*|\mathbf{y}, \mathbf{x}_*) = \mathcal{N}(f^*|\mathbf{m}^T \boldsymbol{\phi}_*, \boldsymbol{\phi}_*^T \mathbf{S} \boldsymbol{\phi}_*)$$

- and the *posterior predictive distribution* for y^* becomes

$$p(y^*|\mathbf{y}, \mathbf{x}_*) = \mathcal{N}(y^*|\mathbf{m}^T \boldsymbol{\phi}_*, \boldsymbol{\phi}_*^T \mathbf{S} \boldsymbol{\phi}_* + \sigma^2)$$

since $\mathbb{V}[\epsilon] = \sigma^2$



Posterior Predictive distributions

Building intuition

- The *posterior distribution* is $p(\mathbf{w}|\mathbf{y}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S})$

- Two *model quantities* of interest

$$f^* = f(\mathbf{x}_*) = \mathbf{w}^T \boldsymbol{\phi}_*$$

$$y^* = y(\mathbf{x}_*) = \mathbf{w}^T \boldsymbol{\phi}_* + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

- Computing the *posterior* mean and variance

$$\mathbb{E}[f^*] = \mathbb{E}[\mathbf{w}^T \boldsymbol{\phi}_*] = \mathbb{E}[\mathbf{w}^T] \boldsymbol{\phi}_* = \mathbf{m}^T \boldsymbol{\phi}_*$$

$$\mathbb{V}[f^*] = \mathbb{V}[\mathbf{w}^T \boldsymbol{\phi}_*] = \boldsymbol{\phi}_*^T \mathbb{V}[\mathbf{w}^T] \boldsymbol{\phi}_* = \boldsymbol{\phi}_*^T \mathbf{S} \boldsymbol{\phi}_*$$

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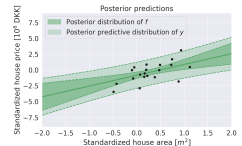
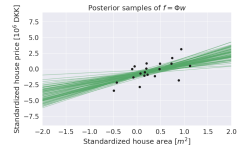
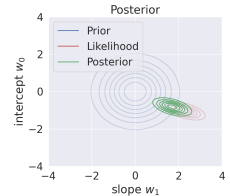
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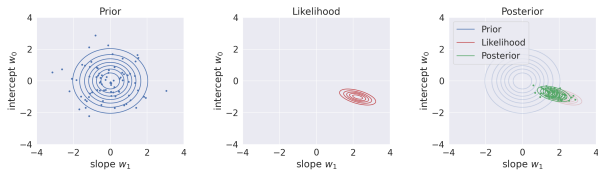
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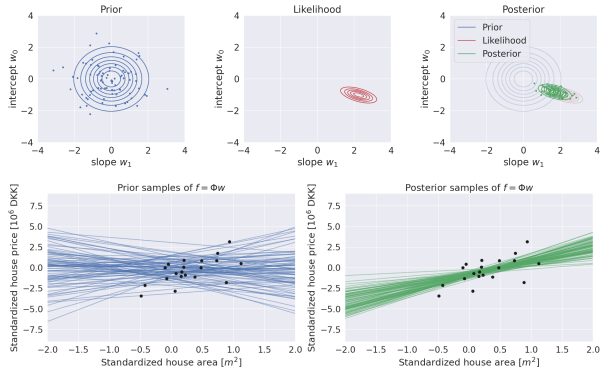
- We can also compute the *prior* predictive distribution.



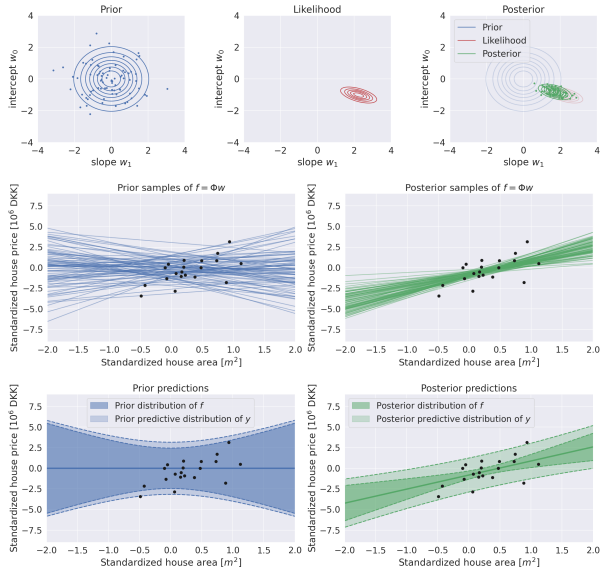
Prior and posterior predictive distributions



Prior and posterior predictive distributions

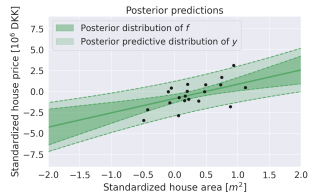
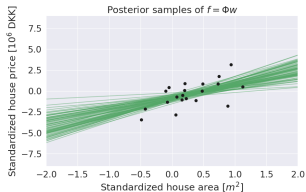


Prior and posterior predictive distributions



Posterior Predictive distributions

Epistemic and aleatoric uncertainty

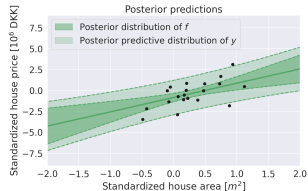
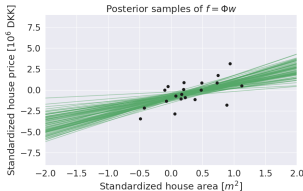


Posterior Predictive distributions

Epistemic and aleatoric uncertainty

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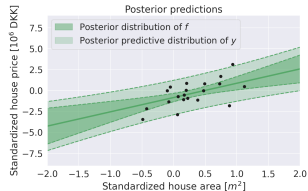
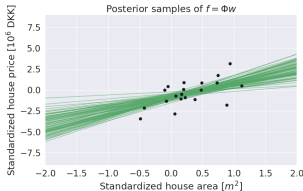
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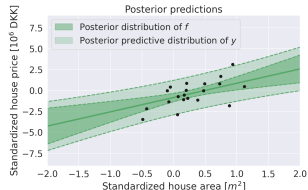
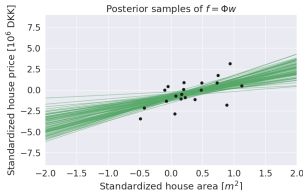
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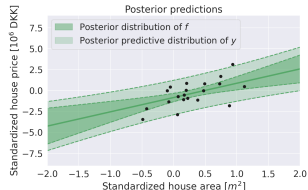
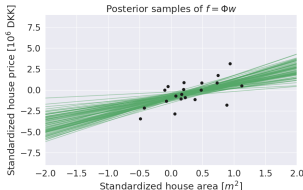
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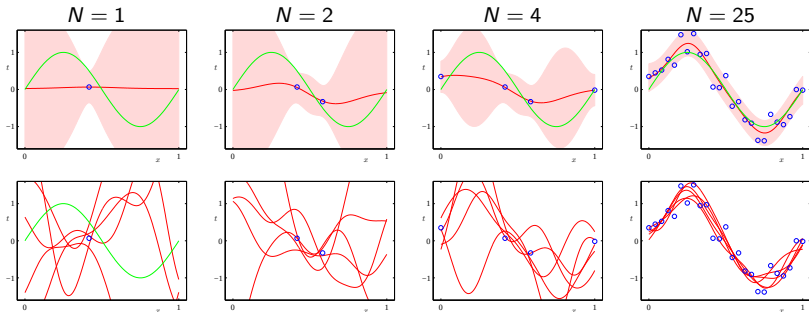
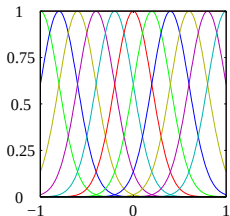
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- The term $\phi(\mathbf{x}_*)^T \mathbf{S} \phi(\mathbf{x}_*)$ is the *posterior uncertainty projected to data space*
- A couple of questions for you:
 1. What happens to the predictive variance when $N \rightarrow \infty$?
 2. Why is the uncertainty "U-shaped"?
 3. Why does the aleatoric uncertainty appear to dominate the total uncertainty near the data?



Example: posterior predictive distributions

- *Predictive distributions* for simple sinoidal toy dataset using Gaussian Basis functions
- Samples from the posterior

Feature extractors $\phi_i(x)$



Dealing with hyperparameters

But what about the hyperparameters?

- The Bayesian linear regression model

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}) \quad (\text{prior})$$

$$p(\mathbf{y} | \mathbf{w}) = \mathcal{N}(\mathbf{y} | \Phi \mathbf{w}, \sigma^2 \mathbf{I}) \quad (\text{likelihood})$$

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- ... and integrate them out

$$p(y^*|\mathbf{y}) = \iiint p(y^*|\mathbf{w}, \beta)p(\mathbf{w}|\mathbf{y}, \alpha, \beta)p(\alpha, \beta|\mathbf{y}) d\mathbf{w}d\alpha d\beta$$

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- ... but this is not analytically tractable. Later in the course we will learn tools to deal with this in general

The evidence approximation

- How to deal with this bastard?

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$$p(\alpha, \beta|\mathbf{y}) \propto p(\mathbf{y}|\alpha, \beta) p(\alpha, \beta) \propto p(\mathbf{y}|\alpha, \beta)$$

- We can estimate $\hat{\alpha}, \hat{\beta}$ by *optimizing the marginal likelihood* $p(\mathbf{y}|\alpha, \beta)$

$$\hat{\alpha}, \hat{\beta} = \arg \max_{\alpha, \beta} \log p(\mathbf{y}|\alpha, \beta)$$

Sinoidal example revisited using the evidence approximation

